MATH221 Midterm Examination, Fall 2025

KU Email: _____

Name (Korean or English): Solutions

Student ID # _____

Date	Tuesday, June 17th
Time Period	1:30-2:45pm
Duration of Test	75 minutes
Number of Test Pages	10 pages (5 double-sided sheets) in total
Total Possible Marks	54
Additional Materials Allowed	None. NO CALCULATORS.

Instructions

1. Answer the questions in the spaces provided. If you require additional space to answer a question, please use the extra page(s) and refer to this page in your solutions.
2. Your grade will be influenced by how clearly you express your ideas, and how well you organize your solutions.
3. **This exam will be scanned and marked electronically.** Be sure to write legibly and **make sure that your writing isn't too faint / light.**
4. DO NOT WRITE NOTES/FORMULAS ON THIS COVER PAGE. If you do, you may get ZERO.
5. Writing any language other than **ENGLISH** on a problem will result **ZERO** on that problem. Writing other languages on a scratch work is okay, but you need to indicate which work is a scratch work and which is not.
6. Convention: $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, vector spaces are assumed to be $\mathbb{F}$-vector spaces, and column vectors can be rewritten as a tuple (not a row vector), i.e.,

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = (a_1, a_2, \ldots, a_n) \neq \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}.$$

1. Compute the following matrix-vector multiplications:

   (a) (6 points)

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{bmatrix} \left(3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \right) = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 \cdot 1 - 2 \\ 3 \cdot 1 - (-1) \\ 3 \cdot 1 - 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \cdot 1 + 4 \cdot 1 + 3(-1) \\ 0 \cdot 1 - 2 \cdot 4 + 1(-1) \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \end{bmatrix}$$

   (b) (6 points)

$$\left(2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 2 & 0 \end{bmatrix} \right) \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + 1 & 2 \cdot 0 + 1 \\ 2 \cdot 0 - 1 & 2 \cdot 1 + 1 \\ 2 \cdot 1 + 2 & 2 \cdot 1 + 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \cdot 2 + 1 \cdot 3 \\ -1 \cdot 2 + 3 \cdot 3 \\ 4 \cdot 2 + 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \\ 14 \end{bmatrix}$$

2. (12 points) Find all values of $t \in \mathbb{R}$ such that the following vectors are linearly dependent in $\mathbb{R}^3$:

$$v_1 = (4, 2, -1), \quad v_2 = (-2, 2, 3), \quad v_3 = (0, 6, t).$$

Sol 1:

v_1, v_2, v_3 lin dep $\iff \exists (x, y, z) \in \mathbb{R}^3 \setminus \{0\}$ s.t. $xv_1 + yv_2 + zv_3 = 0$.

$$x\begin{bmatrix}4\\2\\-1\end{bmatrix} + y\begin{bmatrix}-2\\2\\3\end{bmatrix} + z\begin{bmatrix}0\\6\\t\end{bmatrix} = 0 \iff \begin{bmatrix}4x - 2y + 0\cdot z\\2x + 2y + 6z\\-x + 3y + tz\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \begin{matrix}(1)\\(2)\\(3)\end{matrix}$$

(1): $4x - 2y = 0 \Rightarrow 2y = 4x \Rightarrow y = 2x \Longrightarrow$ (2): $2x + 2(2x) + 6z = 0 \Rightarrow 6x = -6z \Rightarrow z = -x$

$\Rightarrow$ (3): $-x + 3(2x) + t(-x) = 0 \Rightarrow (-t + 5)x = 0 \Rightarrow x = 0$ or $t = 5$.

If $t \neq 5$, then $x = 0 \Rightarrow y = 2 \cdot 0 = 0$ & $z = -x = 0 \Rightarrow v_1, v_2, v_3$ lin indep.

If $t = 5$, then can choose $x = 1 \Rightarrow y = 2 \cdot 1 = 2$ & $z = -x = -1 \Rightarrow v_1 + 2v_2 - v_3 = 0$ lin dep.

Hence, $\boxed{t = 5}$.

Sol 2:

$v_1 \neq 0 \neq v_2$, and $\forall c \in \mathbb{R}$, $cv_1 = v_2 \iff (4c, 2c, -c) = (-2, 2, 3)$

but $4c = -2 \Rightarrow c = -\frac{1}{2}$ & $2\left(-\frac{1}{2}\right) = -1 \neq 2$

$\Rightarrow v_1, v_2$ lin indep.

From class (§1.5), v_1, v_2 lin indep then v_1, v_2, v_3 lin dep $\iff v_3 \in \text{span}(v_1, v_2)$.

Check when $v_3 = xv_1 + yv_2$.

$$x\begin{bmatrix}4\\2\\-1\end{bmatrix} + y\begin{bmatrix}-2\\2\\3\end{bmatrix} = \begin{bmatrix}0\\6\\t\end{bmatrix} \iff \begin{bmatrix}4x - 2y\\2x + 2y\\-x + 3y\end{bmatrix} = \begin{bmatrix}0\\6\\t\end{bmatrix} \begin{matrix}(1)\\(2)\\(3)\end{matrix}$$

(1): $4x - 2y = 0 \Rightarrow 2y = 4x \Rightarrow y = 2x \Rightarrow$ (2): $2x + 2(2x) = 6 \Rightarrow 6x = 6 \Rightarrow x = 1$

$\Rightarrow$ (3): $-1 + 3(2 \cdot 1) = t \Rightarrow \boxed{t = 5}$.

3. (12 points) Consider the following linear transformation:

$$T\colon \mathbb{R}^4 \to \mathbb{R}^3$$
$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \mapsto \begin{bmatrix} x-y \\ 2x-y+z \\ y+z \end{bmatrix}.$$

Find a basis of $\ker(T)$.

$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \ker(T) \iff$ $x-y=0$ (1); $2x-y+z=0$ (2); $y+z=0$ (3).

(1): $x-y=0 \Rightarrow x=y$

$\Rightarrow$ (2): $2x-x+z=0 \Rightarrow x+z=0 \Rightarrow z=-x$

$\Rightarrow$ (3): $x+(-x)=0 \Rightarrow 0=0$.

Thus $\ker(T) = \left\{ \begin{bmatrix} x \\ x \\ -x \\ w \end{bmatrix} : x, w \in \mathbb{R}^2 \right\} = \left\{ x\begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix} + w\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} : x, w \in \mathbb{R} \right\}$

$= \text{span}\left((1,1,-1,0), (0,0,0,1)\right)$.

B/c $(1,1,-1,0) \neq 0 \neq (0,0,0,1)$ and $c(1,1,-1,0) = (c,c,-c,0) \neq (0,0,0,1)$ $\forall c \in \mathbb{R}$,

$\boxed{(1,1,-1,0), (0,0,0,1)}$ is a basis of $\ker(T)$.

4. (9 points) Suppose that V is a vector space and $\{u, v, w, s\}$ is a linearly independent subset of V. Prove that $\{u + s, v + s, w + s\}$ is also a linearly independent subset of V.

If $\{u+s, v+s, w+s\}$ lin dep, then $\exists (a,b,c) \in \mathbb{R}^3 \setminus \{0\}$ s.t.

$a(u+s) + b(v+s) + c(w+s) = 0 \Rightarrow au + bv + cw + (a+b+c)s = 0$

$\Rightarrow a, b, c, a+b+c = 0$ b/c $\{u,v,w,s\}$ lin indep, ※ to $(a,b,c) \neq 0$.

Thus, $\{u+s, v+s, w+s\}$ is lin indep. □

5. (9 points) Suppose that $T\colon \mathbb{R}[x]_{\leq 6} \to M_2(\mathbb{R})$ is a linear map such that $T(x^i) = 0$ for all $i = 3, 4, 5, 6$. Prove that $\operatorname{im}(T) \neq M_2(\mathbb{R})$.

Hint: Compare dimensions of $\operatorname{im}(T)$ and $M_2(\mathbb{R})$.

Sol 1: $x^i \in \ker(T) \quad \forall\, 3 \leq i \leq 6 \Rightarrow \operatorname{span}(x^3, x^4, x^5, x^6) \subseteq \ker(T)$.

x^3, x^4, x^5, x^6 lin indep (b/c subset of std basis of $\mathbb{R}[x]_{\leq 6}$)

$\Rightarrow 4 = \dim \operatorname{span}(x^3, x^4, x^5, x^6) \leq \text{nullity}(T)$

$\Rightarrow \operatorname{rank}(T) = \dim \mathbb{R}[x]_{\leq 6} - \text{nullity}(T) \leq 7 - 4 = 3$ by R-N Thm.

As $\dim M_2(\mathbb{R}) = 2 \cdot 2 = 4 > 3 \geq \operatorname{rank}(T)$, $\operatorname{im}(T) \subsetneq M_2(\mathbb{R})$. □

Sol 2: $x^0, x^1, \ldots, x^6$ std basis of $\mathbb{R}[x]_{\leq 6}$

$\Rightarrow \operatorname{im}(T) = \operatorname{span}(T(x^0), T(x^1), T(x^2), T(x^3), \ldots, T(x^6))$

$= \operatorname{span}(\underbrace{T(x^0), T(x^1), T(x^2)}_{3 \text{ elts}})$ b/c $T(x^i) = 0 \quad \forall\, 3 \leq i \leq 6$.

$\Rightarrow \operatorname{rank}(T) \leq 3 < 4 = 2 \cdot 2 = \dim M_2(\mathbb{R}) \Rightarrow \operatorname{im}(T) \neq M_2(\mathbb{R})$. □

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